KRISHI-DRISHTI (■■■■-■■■■■■)

Date-Wise Field Survey & Empirical Prototype Validation Report

An Offline-First Handheld Optical Wand and Microclimate Gating System for Solanaceae Foliar Disease Screening

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Mentor Teacher: Soni Kumari (PGT Science)	**Timeline:** August 3, 2026 – November 12, 2026
Target Region: Begusarai (Barauni & Teghra blocks), Bihar	**Standardized BOM:** ■1,480 (Strictly under ■1,500)

1. EXECUTIVE SUMMARY & RECONCILED FIVE-PHASE TIMELINE

Between **August 3, 2026, and November 12, 2026**, an empirical investigation was completed to develop and evaluate KRISHI-DRISHTI—an ultra-low-cost (■1,480 BOM) handheld optical wand and edge-AI diagnostic assistant for smallholder tomato and potato growers in rural Bihar.

• **Phase I (Aug 3 – Aug 20):** Baseline survey of 30 vegetable smallholders in Begusarai documenting chemical costs (~■2,150/acre/spray), absence of pathology labs, and in-field cellular dropouts (80%).

• **Phase II (Aug 25 – Sep 28):** Dual-ESP32 prototyping, 8cm optical shroud calibration, SSD1306 OLED (GPIO 21/22), DHT11 (GPIO 18), and 470uF anti-brownout stabilization.

• **Phase III (Oct 1 – Oct 26):** MobileNetV3 Float16 edge quantization (2.01MB) trained on 7,710 records (6,965 unique images) with 171-hash master blacklist. 60-specimen botanical benchmark achieved 88.33% strict and 92.98% clinical accuracy.

• **Phase IV (Oct 28 – Nov 8):** Unconstrained camera testing discovered specular shadow failure modes (16.7% raw accuracy under torch), validating the physical necessity of the 8cm shroud and DHT11 microclimate gating (suppressing Late Blight spray advice when RH < 85%). Usability demos conducted with 5 local farmers (15 test scans).

• **Phase V (Nov 9 – Nov 12):** Firmware hardening (v2.3), ■1,480 BOM audit, and final exhibition documentation sign-off.

2. PHASE I: SOCIO-ECONOMIC SURVEY FINDINGS (N = 30 SMALLHOLDERS)

Investigation Metric	Pre-Intervention Baseline (N = 30 Farmers)	KRISHI-DRISHTI Prototype Capability
Diagnostic Method	86.7% (26/30) relied on guesswork or dealer recommendations	Instant objective 420 ms foliar screening on standing plants
Cellular Internet Access	80.0% (24/30) had no reliable mobile internet in open crop fields	100% offline edge inference on local hardware without data packets
Chemical Spray Waste	Averaged ■2,150/acre per spray on broad-spectrum chemical cocktails	Prevents 2–3 redundant sprays, saving ■4,300–■6,800/acre annually
Diagnostic Access Time	7–10 days if sending physical samples to distant district labs	Immediate on-field screening directly in the crop row

3. PHASE III: CONTROLLED BOTANICAL BENCHMARK (N = 60 FRESH SPECIMENS)

Tested with 8 cm focal shroud and circular matte acrylic diffuser disc: strict 6-class exact accuracy reached **88.33% (53 / 60)**; pathogen-family agreement reached **90.00% (54 / 60)**. Shannon Entropy gating ($H > 1.10$) safely intercepted 3 ambiguous samples, raising clinical agreement on accepted specimens to **92.98% (53 / 57)** with an average latency of 420 ms.

4. PHASE IV: CAMERA AUDIT & SHADOW FAILURE MODE DISCOVERY

Auditing unconstrained mobile cameras under artificial night illumination and public benchmarks (PlantDoc) revealed raw accuracy falling to 33.3% and healthy leaves misdiagnosed as Late Blight (16.7% raw accuracy) due to specular torch reflections and steep drop-shadows mimicking necrotic rings. **This finding proved why KRISHI-DRISHTI's multi-modal physical design is essential:**

• **Mechanical Fix:** The physical 8 cm tubular shroud and circular matte acrylic diffuser eliminate direct point-source glare and radial shadows.

• **Microclimate Gating:** The DHT11 sensor tracks relative humidity. Because *Phytophthora infestans* cannot sporulate below 85% RH, the controller wand automatically suppresses chemical Late Blight recommendations if ambient humidity is low, regardless of CNN output.

• **Software Entropy Gating:** Ambiguous distributions (H > 1.10) output UNCERTAIN: Rescan.

5. STANDARDIZED BILL OF MATERIALS (BOM: ■1,480)

Subsystem / Component	Technical Specification	Operational Function in Prototype	Unit Cost
ESP32-CAM Node	Dual-core 240MHz, 4MB PSRAM, OV2640 installed	Image capture, SoftAP web server, SD logger	■540
ESP32 DevKit V1	Dual-core 240MHz, 30-pin, ESP-NOW link	Master controller, button logic, OLED driver	■420
SSD1306 OLED Display	0.96 inch, 128x64 px, I2C (GPIO 21/22)	4-page status and telemetry carousel	■190
MicroSD Card (8 GB)	Class 10, FAT32 format, 1-bit SD_MMC mode	Image storage and offline WebAssembly assets	■160
DHT11 Sensor	Temp 0-50 C, Humidity 20-90% RH (GPIO 18)	Microclimate disease-triangle risk tracking	■95
Optical Shroud & Diffuser	8 cm focal tube, matte acrylic diffuser disc	Standardizes focal distance and eliminates glare	■45
Passives & Switches	Tactile switches (GPIO 26/27), 470uF cap, wiring	Debounced buttons and power rail protection	■30
Total Bill of Materials	**100% Commercial Off-The-Shelf Hardware**	**Complete Handheld Prototype**	**■1,480**

Official Verification: *We certify that all survey data, technical specifications, and empirical records in this report represent authentic student research conducted under teacher guidance at Udaan International School, Begusarai. Date: 12 November 2026.*